



(globals)

Exp 8.cpp Exp 9.cpp Exp 10.cpp Exp 11.cpp Exp 12.cpp Exp 13.cpp [*] Exp 14.cpp Exp 15.cpp Exp 16.cpp Exp 17.cpp Exp 18.cpp

```
1 #include <stdio.h>
2 #include <stdlib.h>
3 #include <windows.h>
4 #define BUFFER_SIZE 5
5 #define MAX_ITEMS 10
6 int buffer[BUFFER_SIZE];
7 HANDLE sem_empty, sem_full;
8 HANDLE buffer_mutex;
9 int produced_items = 0, consumed_items = 0;
10 DWORD WINAPI producer(LPVOID arg) {
11     while (1) {
12         WaitForSingleObject(buffer_mutex, INFINITE);
13         if (produced_items >= MAX_ITEMS) {
14             ReleaseMutex(buffer_mutex);
15             break;
16         }
17         ReleaseMutex(buffer_mutex);
18         WaitForSingleObject(sem_empty, INFINITE);
19         WaitForSingleObject(buffer_mutex, INFINITE);
20         for (int i = 0; i < BUFFER_SIZE; ++i) {
21             if (buffer[i] == 0) {
22                 buffer[i] = produced_items + 1;
23                 printf("Produced: %d\n", buffer[i]);
24                 produced_items++;
25                 break;
26             }
27         }
28     }
29 }
```

Resources Compile Log Debug Find Results Close

Compilation results...

- Errors: 0
- Warnings: 0
- Output Filename: C:\AADHITHYA\Exp18.exe
- Output Size: 132.173828125 KiB
- Compilation Time: 2.58s

(globals)

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```
49         break;
50     }
51 }
52 ReleaseMutex(buffer_mutex);
53 ReleaseSemaphore(sem_empty, 1, NULL);
54 Sleep(400);
55 }
56 return 0;
57 }
58 int main(void) {
59     HANDLE producer_thread, consumer_thread;
60     sem_empty = CreateSemaphore(NULL, BUFFER_SIZE, BUFFER_SIZE, NULL);
61     sem_full = CreateSemaphore(NULL, 0, BUFFER_SIZE, NULL);
62     buffer_mutex = CreateMutex(NULL, FALSE, NULL);
63     if (sem_empty == NULL || sem_full == NULL || buffer_mutex == NULL) {
64         printf("Error initializing synchronization objects.\n");
65         return 1;
66     }
67     for (int i = 0; i < BUFFER_SIZE; i++) buffer[i] = 0;
68     producer_thread = CreateThread(NULL, 0, producer, NULL, 0, NULL);
69     consumer_thread = CreateThread(NULL, 0, consumer, NULL, 0, NULL);
70     WaitForSingleObject(producer_thread, INFINITE);
71     WaitForSingleObject(consumer_thread, INFINITE);
72     CloseHandle(sem_empty);
73     CloseHandle(sem_full);
```

sources

Compile Log

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```
56     return 0;
57 }
58 int main(void) {
59     HANDLE producer_thread, consumer_thread;
60     sem_empty = CreateSemaphore(NULL, BUFFER_SIZE, BUFFER_SIZE, NULL);
61     sem_full = CreateSemaphore(NULL, 0, BUFFER_SIZE, NULL);
62     buffer_mutex = CreateMutex(NULL, FALSE, NULL);
63     if (sem_empty == NULL || sem_full == NULL || buffer_mutex == NULL) {
64         printf("Error initializing synchronization objects.\n");
65         return 1;
66     }
67     for (int i = 0; i < BUFFER_SIZE; i++) buffer[i] = 0;
68     producer_thread = CreateThread(NULL, 0, producer, NULL, 0, NULL);
69     consumer_thread = CreateThread(NULL, 0, consumer, NULL, 0, NULL);
70     WaitForSingleObject(producer_thread, INFINITE);
71     WaitForSingleObject(consumer_thread, INFINITE);
72     CloseHandle(sem_empty);
73     CloseHandle(sem_full);
74     CloseHandle(buffer_mutex);
75     CloseHandle(producer_thread);
76     CloseHandle(consumer_thread);
77
78     printf("\nAll items produced and consumed successfully.\n");
79     return 0;
80 }
```

Resources Compile Log Debug Find Results Close

Compilation results...

- Errors: 0
- Warnings: 0
- Output Filename: C:\AADHITHYA\Exp18.exe
- Output Size: 132.173828125 KiB
- Compilation Time: 2.50s

```
C:\AADHITHYA\Exp18.exe
Produced: 1
Consumed: 1
Produced: 2
Produced: 3
Consumed: 2
Produced: 4
Consumed: 4
Produced: 5
Produced: 6
Consumed: 5
Produced: 7
Produced: 8
Consumed: 7
Produced: 9
Produced: 10
Consumed: 9
Consumed: 3
Consumed: 6
Consumed: 8
Consumed: 10

All items produced and consumed successfully.

-----
Process exited after 4.26 seconds with return value 0
Press any key to continue . . .

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- Output filename: C:\AADHITHYA\Exp18.exe
- Output Size: 132.173628125 KiB
- Compilation Time: 2.58s
```